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CS5100 Final Project Writeup

Introduction

As a volleyball player who has not always received hands-on coaching, I would watch film of my own play and recognize issues with my form but struggle to pinpoint exactly what to improve. This experience inspired me to build a volleyball-specific tool that uses computer vision to extract key biomechanical metrics from a hitter's swing and provide concrete, actionable feedback.

Current computer vision tools applied to volleyball focus primarily on ball tracking, highlight detection, and skipping through downtime in game footage. Individual biomechanical form analysis remains largely inaccessible. Tools that provide this level of feedback typically require expensive motion capture equipment, professional coaching, or laboratory settings. For example, (Serrien et al., 2016) used 6 synchronized 250hz cameras with reflective markers on the athletes bodies to extract the 3D kinematics of volleyball spiking, equipment far beyond what is available to most.

While recent advances in markerless pose estimation offer a more accessible alternative, standard models perform poorly on the fast, dynamic movements characteristic of athletic performance, meaning off-the-shelf solutions are insufficient for this problem without careful adaptation. A computer vision based form analysis tool accessible to everyday players is therefore something the volleyball world is currently missing, making this both a meaningful and technically interesting challenge.

Methods

Most athletes simply use their phones to record their games, myself included. The most common recording angle is from the back due to the visibility of the court and ease of setup. This back-facing angle informed both the design of my system and its limitations. While it provides a clear view of trunk and hip rotation, it obscures movements like the arm swing. For my project, this is the angle I used for testing and required for users.

Given this constraint, I built a pipeline using MediaPipe's Pose Landmarker (heavy model) to extract 33 skeletal keypoints from each video frame, then computed biomechanical metrics from those keypoints and compared them against benchmarks derived from sports science research (Serrien et al., 2016). The pipeline was implemented in Python using OpenCV for video processing, NumPy and SciPy for signal processing, and Matplotlib for visualization.

When running pose estimation on the full video frame, MediaPipe would frequently lock onto the wrong person, typically whoever had the highest detection confidence rather than the intended hitter. Testing on manually cropped videos had significantly more accurate landmark placement, motivating a dynamic cropping approach. At the start of each session, the user draws a bounding box around the hitter to define the initial region of interest (ROI). Each frame is cropped to this region before being passed to MediaPipe, ensuring the model focuses exclusively on the hitter. The ROI center updates each frame based on the detected torso position, keeping the crop centered on the hitter as they move across the court. Landmark coordinates returned by MediaPipe are then translated back to full frame pixel space for display.

From each frame's skeletal keypoints, I computed three lengths: torso length (shoulder midpoint to hip midpoint), shoulder line length (left to right shoulder), and hip line length (left to

right hip). As the hitter moves away from the camera, perspective causes all projected lengths to shrink uniformly. To account for this, I normalized shoulder and hip lengths as a ratio against torso length, since both shrink at the same rate with camera distance, the ratio remains stable and reflects only rotational changes. The frame where each ratio is maximized corresponds to when the hips or shoulders are most square to the camera and net. The rotation angle at any frame is then: angle = arccos(observed_length / max_length).

Angular velocities were computed by applying a Savitzky-Golay smoothing filter (window length 9, polynomial order 2) to the angle time series, then taking the numerical gradient via NumPy. Raw finite differences were too noisy at 30fps to be useful, smoothing was essential before differentiation. The resulting shoulder angular velocity curve closely matched the shape of the trunk rotation velocity curves reported by Serrien et al. (2016), validating the approach.

Four metrics were extracted from the window (400ms before to 80ms after contact, following Serrien et al., 2016):

- **Peak trunk rotation velocity**: maximum shoulder angular velocity in °/s
- **Peak timing**: how many milliseconds before contact peak velocity occurs
- **Onset**: when shoulder angular velocity first becomes consistently positive, representing the start of forward trunk rotation
- **Hip-shoulder peak timing difference**: how many milliseconds earlier the hip reaches peak velocity compared to the shoulder, quantifying kinetic chain sequencing

These metrics were compared against benchmarks from Serrien et al. (2016) for junior elite and professional male and female players. Session data is stored in a JSON file alongside

the video path, date, and gender, allowing trend analysis across multiple sessions using linear regression on each metric over time.

Results

To evaluate the system, I tested it across five of my own recorded volleyball swings with varying quality and filming conditions, including both 30fps and 60fps footage. The results revealed both the capabilities and limitations of the approach.

The dynamic cropping approach proved effective in controlled filming conditions where the hitter was clearly isolated as the primary subject. However, in game footage with multiple players, overlapping players caused the ROI to occasionally drift to a nearby player, particularly when the hitter and a blocker overlapped near the net. Once the ROI locked onto the wrong person it would continue tracking them until they separated as the hitter was in movement while the ROI would stay locked onto the same person , corrupting the rest of the landmark data. This was most problematic in the final frames approaching contact, the window most important for analysis. For the purposes of this project, users are advised to film in controlled settings where the hitter is the primary subject of the frame.

MediaPipe's landmark placement was generally reliable during the approach and wind-up phases but degraded significantly during the arm swing and contact moment due to motion blur and the unusual body positions involved. Occasional landmark teleportation where a keypoint would suddenly jump to an anatomically incorrect position for one or two frames introduced spikes in the computed angular velocities. The Savitzky-Golay smoothing filter mitigated much of this noise, but extreme outliers could still inflate peak velocity readings. For example, peak

trunk rotation velocity across my five test sessions ranged from 162°/s to 465°/s, with the higher values likely inflated by landmark noise rather than reflecting true rotational speed.

Across the recorded sessions, the average metrics were:

- Peak trunk rotation velocity: 362.9°/s (benchmark: ~400°/s)
- Peak timing before contact: 161.9ms (benchmark: 60-80ms)
- Onset before contact: 291.2ms (benchmark: 180-220ms)

Here is an example graph of hip and shoulder velocity calculated from my program compared to the graph of torso velocity in the Serrien et al Paper (2016)

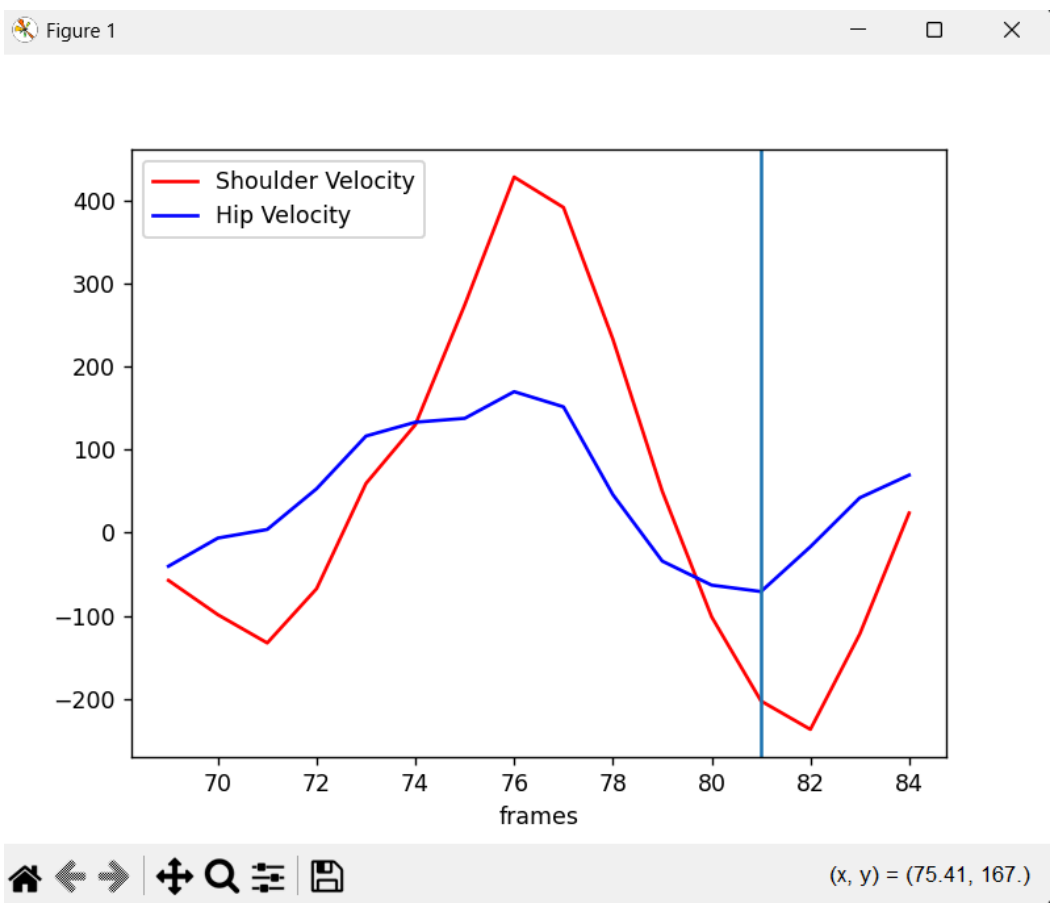


Figure 1: Hip and shoulder angular velocity extracted by this system for a sample session.

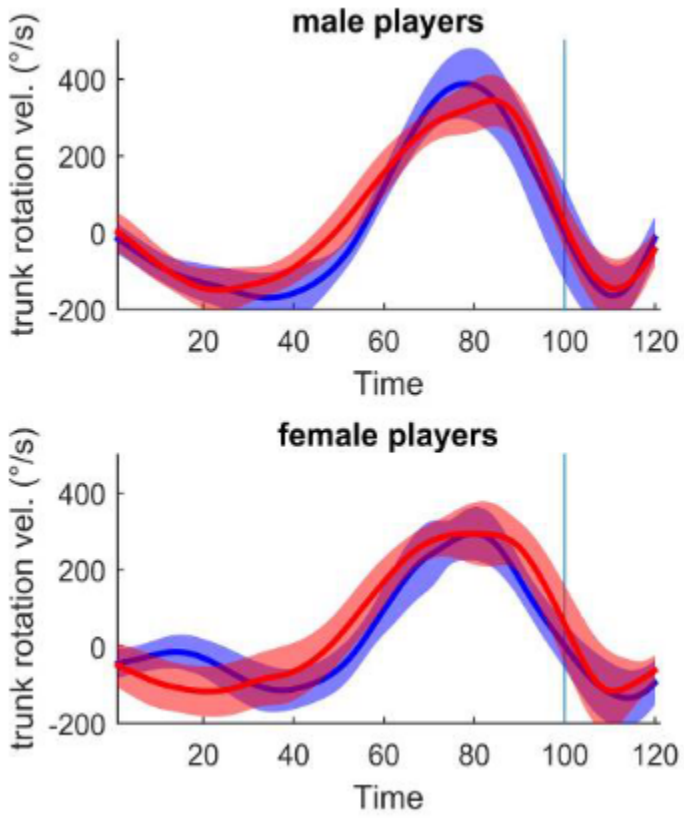


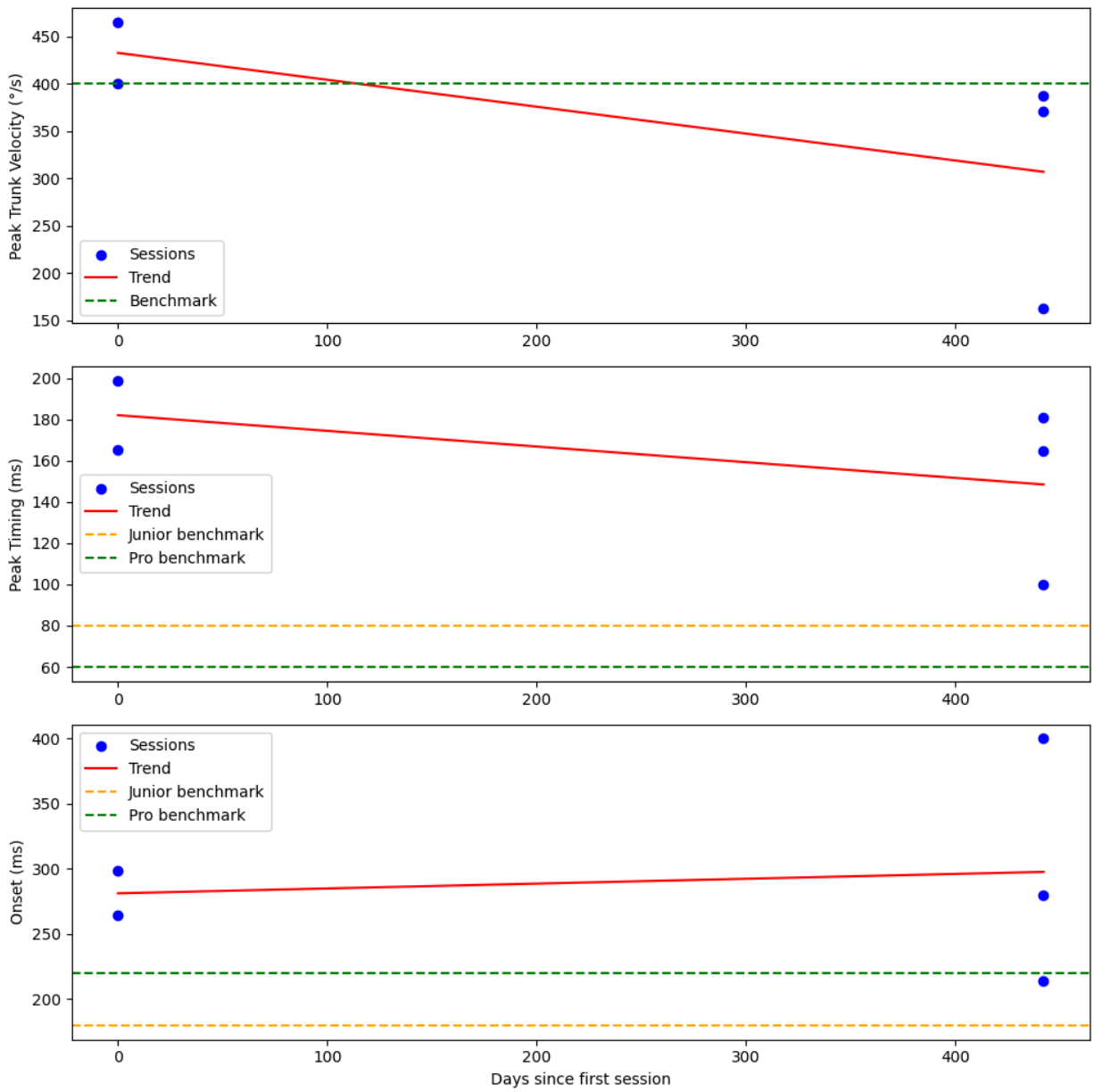
Figure 2: Figure 1: Trunk rotation velocity over time for top level (red) and junior elite (blue) volleyball players, reproduced from Serrien et al. (2016). Positive values indicate rotation toward the net.

My peak trunk rotation velocity was close to the elite benchmark, which is encouraging given the limitations of 30fps smartphone footage. The timing metrics showed more deviation peak velocity occurred roughly 100ms later than elite benchmarks and onset was approximately 100ms later as well. This could reflect genuine technique differences, contact frame tagging imprecision at 30fps (±33ms per frame), or the smoothing filter introducing lag in the velocity

signal. Testing with 60fps footage produced a peak timing of 100ms before contact, closer to but still above the benchmark, suggesting frame rate is a significant factor in timing metric accuracy.

The system's benchmarks are derived from the research article written by Serrien et al (2016), which measured technique specifically for outside attacks, which involve a full approach, jump, and arm swing. Testing on a quick middle attack produced a peak trunk velocity of 162°/s, well below the benchmark of 400°/s. In volleyball, swing technique varies a lot from player to player, some use the more modern circular swing, others stick with the traditional high/low elbow. The system I built, while mainly looking at hip and shoulder velocity, removes some of these possible inaccuracies because no matter what arm technique, torso rotation and hip shoulder separation are key to generating more power and thus hitting harder. Even still due to the inconsistent nature of the sport, there will be swings flagged as needing improvement while actually being the most optimal.

Across sessions, my trunk velocity showed a declining trend of -0.3°/s per day, which is likely an artifact of inconsistent filming conditions and landmark quality across videos rather than genuine performance decline. Timing metrics showed slight improvement (negative trend) which is the correct direction. Meaningful trend analysis will require more sessions filmed under consistent conditions. Below are the graphs of the trends over time that I extracted from my 5 test videos.



Conclusions

Volleyball form analysis, for most non professional players, is typically done subjectively rather than quantitatively simply due to the difficulty of obtaining precise measurements. Professional athletes have access to expensive cameras and simulation technology to visualize their spikes in 3D. For recreational volleyball players this level of analysis is inaccessible. Most

people need to go find more experienced players or coaches to help them which takes time and effort.

This project's approach to the problem, finding key metrics from the most common filming angle using recordings from a smartphone, makes analyzing volleyball much easier and more accessible. The metric calculation from controlled videos produced results that aligned reasonably well with elite benchmarks and captured meaningful biomechanical patterns, particularly the shape of the trunk rotation velocity curve which closely matched the profiles reported by Serrien et al. (2016). Even in some real gameplay footage, with careful ROI selection, meaningful metrics were extractable.

However, compared to the reliability of a multi camera laboratory approach, the single back facing camera brings many fundamental limitations. Depth-plane movements such as the shoulders internal rotation and elbow extension are not measurable and require many assumptions to even predict. The typical 30fps limits precise timing metrics precision and causes more noisy results. The reliance on MediaPipes general purpose pose landmarker, rather than a more specialized sports specific model, also introduces landmark inaccuracies during quick athletic movements. Sports-specific pose estimation datasets such as AthletePose3D (Yeung et al., 2025) show promise for improving accuracy on athletic movements but are not yet trained on volleyball-specific data.

Further improvements to the program include better user experience through a web or mobile interface to reduce friction around video uploading, more automated outlier rejection to reduce the need for users to manually retry sessions, and extraction of additional metrics such as lateral trunk tilt and elbow angle. With more session data, machine learning approaches such as k-means clustering could identify swing archetypes and provide more personalized feedback

beyond simple benchmark comparison and linear regression, a feature planned but not yet feasible with my current small dataset.

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